LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Simple inertial navigation system block diagram.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Simple inertial navigation system block diagram.
5. Specific Objectives	14.1.1.1 Trainee will be able to explain and define a position on the earth.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of INS Block Diagram components using PPT slides; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an Inertial Navigation System?
(b) Define the term 'inertial' in the context of navigation.
(c) List three basic parameters required to define a position on the earth.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Simple INS • Core components of INS • Function of Accelerometers • Function of Gyroscopes • Integration process basics	L & Ds	PPT, IB	15 min
(b)	• Interpretation of Block Diagram • Signal flow analysis • Coordinate transformation • Position calculation logic • Velocity to position integration	L & Ds	PPT, PDF	20 min
(c)	• Earth position definition • Latitude and Longitude • Altitude measurement • Reference frame alignment • System initialization concept	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the INS block diagram converts acceleration into position.
(b) Describe the role of the gyroscope in maintaining the reference frame within the block diagram.

12. Home Assignment and Follow Up	Draw a simple block diagram of an INS and label the input sensors and output position coordinates, preparing to discuss error sources in the next lesson.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Different errors in INS., initial leveling error, initial tilt error, accelerometer error, integrator error, initial azimuth misalignment, azimuth gyro drift, and leveling gyro drift.	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min